

Explainable Network Verification via Subspecifications

User Study - Introduction of Background Knowledge

Network Operations and Network Verifications



For network operators, **aligning** network specifications (network routing policies or network design intents) with the actual device configurations is **an extremely challenging task**.



Network verifier ensures that device configurations **faithfully implement** the network specifications (network routing policies or network design intents) by using formal methods, providing **mathematically correctness guarantees**.

- **Reachability:**

10.0.0.0/8, ISP1, Customer1 $\rightarrow$ R3 $\rightarrow$ R1 $\rightarrow$ ISP1

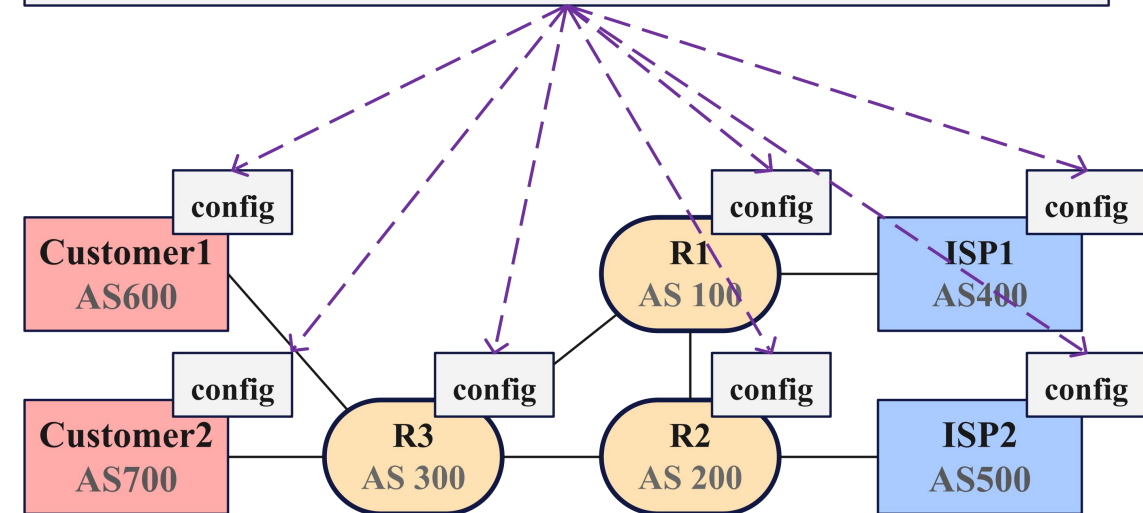
- **Isolation:**

10.0.0.0/8, ISP1, Customer2 $\rightarrow$ ✗

- **Load balance (ECMP, Equal-Cost Multi-Path)**

192.168.0.0/16, ISP1, Customer1 $\rightarrow$ R3 $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ ISP2

192.168.0.0/16, ISP1, Customer1 $\rightarrow$ R3 $\rightarrow$ R2 $\rightarrow$ ISP2



Background: Explainable Network Verification

Network verifiers often give **YES/NO** (*with a counterexample*) answers, without explaining *why*.

Example: suppose we want a BGP policy blocking the private prefix 192.168.0.0/16.

```
1 ip prefix-list private_ips_1 seq 5 permit 192.168.0.0/12 ge 12
2 !
3 ip prefix-list private_ips_2 seq 5 permit 192.168.0.0/16 ge 16
```

Both of them pass the verification of the network verifier.

However, **private_ips_2** is more **precise** than **private_ips_1**.

Overly restrictive filter may block more prefix than necessary.

Explainable Network Verification

Why does a specific configuration field or line satisfy **the network specification** ——
which blocks the private prefix 192.168.0.0/16?

permit AND prefix cover 192.168.0.0/16 AND
prefix ge range = 0~16 AND
prefix ge range >= prefix length

permit

prefix cover 192.168.0.0/16

```
1 ip prefix-list private_ips_1 seq 5 permit 192.168.0.0/12 ge 12
2 !
3 ip prefix-list private_ips_2 seq 5 permit 192.168.0.0/16 ge 16
```

prefix ge range =
12~16

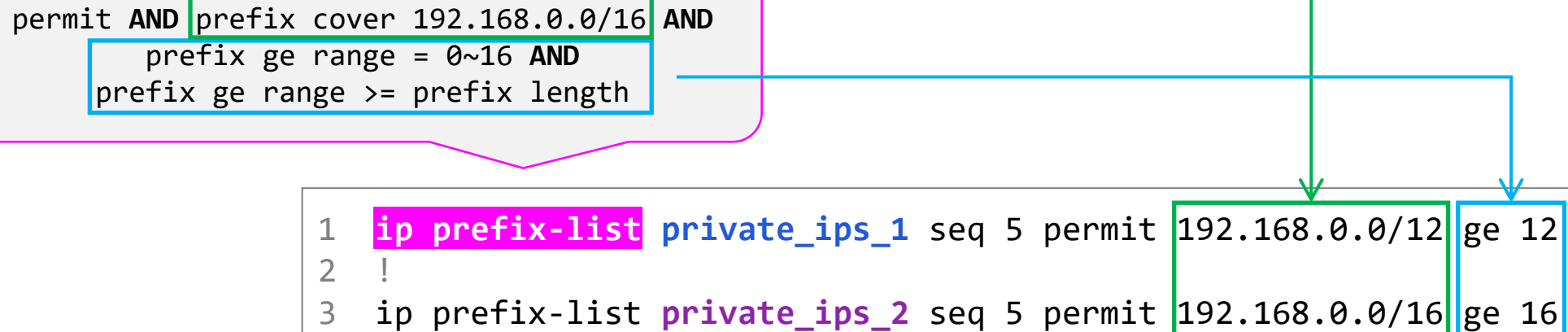
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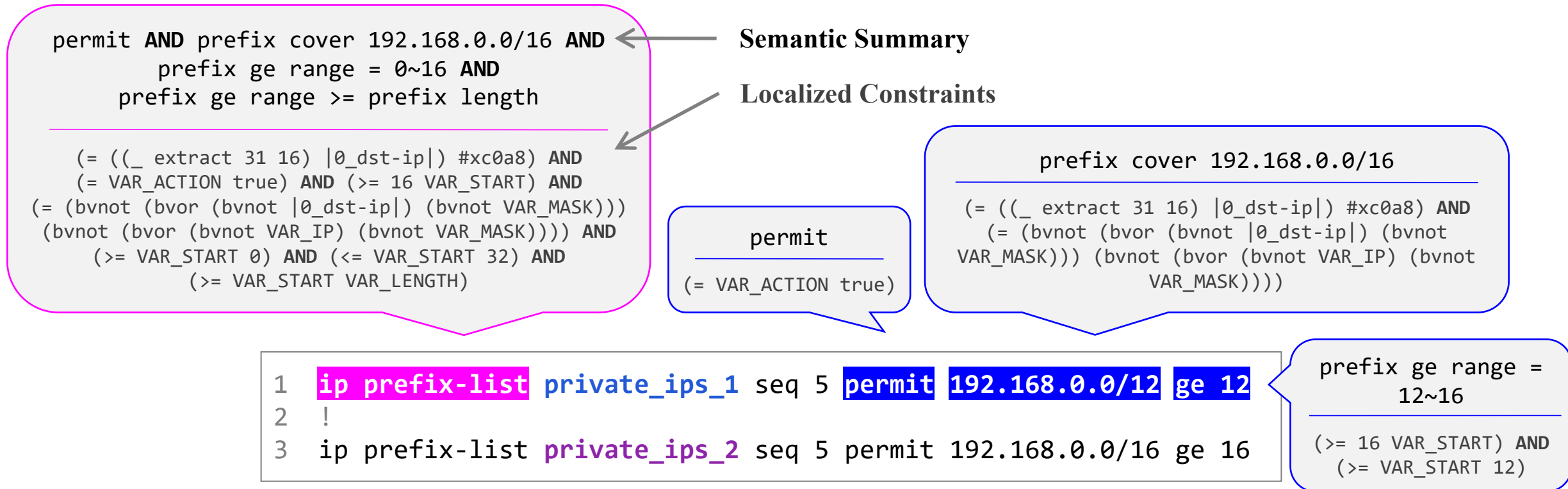


Both of them pass the verification of the network verifier.

private_ips_2 is the more precise option.

Explainable Network Verification via Subspecifications

Localized Subspecifications (Subspecs): the safe modification scope of a configuration field or line that preserves the prior verification success of the network specification — which blocks the private prefix 192.168.0.0/16.



Case 1: Modify a **Single** Configuration Field

The previously verified network specification: blocking the private prefix **192.168.0.0/16**.

- Modification **within** subspec: preserves the verified specification.

prefix ge range = 12~16
(>= 16 VAR_START) AND
(>= VAR_START 12)

1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 14



- Modification **out of** subspec: may violate the verified specification. *(sound but not complete)*

prefix cover 192.168.0.0/16
(= ((_ extract 31 16) |0_dst-ip|) #xc0a8) AND
(= (bvnot (bvor (bvnot |0_dst-ip|) (bvnot VAR_MASK)))
(bvnot (bvor (bvnot VAR_IP) (bvnot VAR_MASK))))

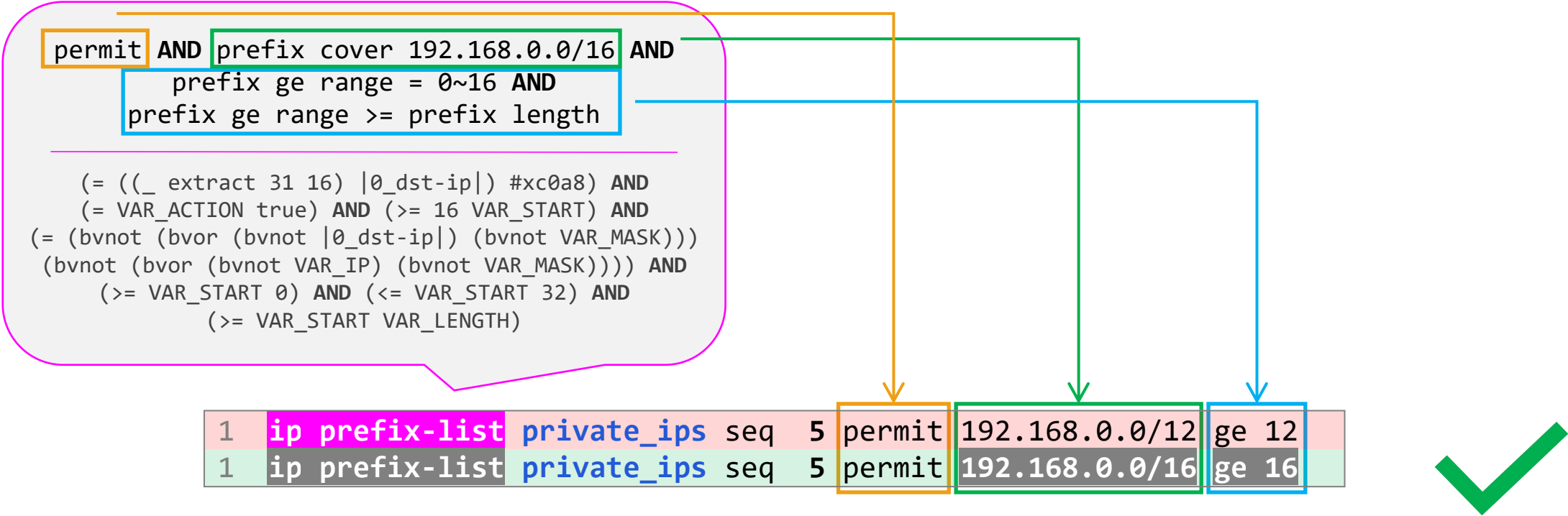
1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq	5	permit	172.16.0.0/12	ge 12



Case 2: Modify Multiple Configuration Fields

The previously verified network specification: blocking the private prefix 192.168.0.0/16.

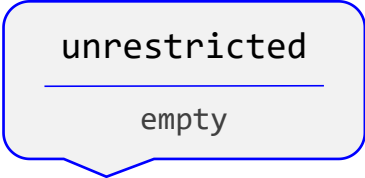
- Modification multiple configuration fields within subspec: preserves the verified specification.



Case 3: Modify a **Single** Configuration Field (Empty Subspec)

The previously verified network specification: blocking the private prefix **10.0.0.0/8**.

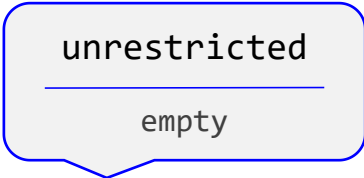
■ **Arbitrary modification:** preserves the verified specification.



1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 14
2	ip prefix-list	private_ips	seq	10	permit	10.0.0.0/8	ge 8



■ **Arbitrary modification:** preserves the verified specification.



1	ip prefix-list	private_ips	seq	5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq	5	permit	172.16.0.0/12	ge 12
2	ip prefix-list	private_ips	seq	10	permit	10.0.0.0/8	ge 8



Case 4: Modify **Multiple** Configuration Fields (Empty Line-level Subspec)

The previously verified network specification: blocking the private prefix **10.0.0.0/8**.

- **Arbitrary modification** multiple configuration fields: preserves the verified specification.

unrestricted

empty

1	ip prefix-list	private_ips	seq 5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq 5	deny	172.16.0.0/12	ge 14
2	ip prefix-list	private_ips	seq 10	permit	10.0.0.0/8	ge 8



- **Safe removal** the entrie configuration line: preserves the verified specification.

unrestricted

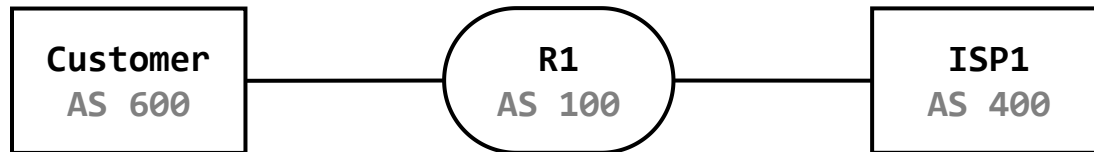
empty

1	ip prefix-list	private_ips	seq 5	permit	192.168.0.0/12	ge 12
1	ip prefix-list	private_ips	seq 5	permit	192.168.0.0/12	ge 12
2	ip prefix-list	private_ips	seq 10	permit	10.0.0.0/8	ge 8



Tips for User Study

- In this user study, the eBGP best route selection process only involves **AS-path length**.
- In this user study, the route-map naming rule is **Router_Direction(IN_FROM/OUT_TO)_Peer**.
 - R1_IN_FROM_ISP1 — BGP import policy on R1 from ISP1 ($R1 \leftarrow ISP1$).
 - R1_OUT_TO_Customer — BGP export policy on R1 to Customer ($R1 \rightarrow Customer$).



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User Study - Introduction of Background Knowledge

Thank you for participating in this user study!